

	Case Name: Biofuel Refinery	Sector	Engineering	
	OR-AS Operations Research - Applications and Solutions www.or-as.be info@or-as.be	Baseline Schedule	Schedule with resources	
Submitted by	Stijn Van Auwenis	Risk Analysis	Schedule with costs	
Date	2016		Random simulation	
File Name	C2016-10 Biofuel Refinery	Project Control	One of nine std. scenarios	
			User defined distributions	
			Automatic tracking	
			Tracking based on user input	

1. Project description

Project authenticity

The construction and commissioning of a biofuel refinery facility. The project consists of activities such as site preparation, engineering design, equipment installation, and system testing, with associated resource and cost data obtained directly from the actual project owner.

2. Project properties

2.1. Baseline Schedule

General		Network topology	
# Activities	24	Serial/Parallel (SP)	18%
Planned Duration (PD)	360	Activity Distribution (AD)	22%
Budget At Completion (BAC)	14,362,625 €	Length of Arcs (LA)	6%
Renewable Resources	-	Topological Float (TF)	21%
Consumable Resources	-		

* standard eight-hour working days

2.2. Risk Analysis

Random simulation by ProTrack was performed using the default symmetric triangular risk distribution profiles.

	Cost sensitivity		
	avg [%]	std dev [%]	skew [-]
CRI-r	16	18	1.87
CRI-rho	29	20	0.59
CRI-tau	37	40	0.94

	Resource sensitivity		
	avg [%]	std dev [%]	skew [-]
CRI-r	NA	NA	NA
CRI-rho	NA	NA	NA
CRI-tau	NA	NA	NA

	Time sensitivity		
	avg [%]	std dev [%]	skew [-]
CI	46	47	0.09
SI	84	21	-2.00
SSI	15	27	1.71
CRI-r	9	6	0.93
CRI-rho	9	7	0.86
CRI-tau	13	8	0.20

2.3. Project Control

2.3.1. Simulated forecasting accuracy

The accuracy of time and cost forecasting methods has been evaluated based on Monte Carlo simulation runs using the risk profiles described in section “2.2. Risk Analysis”. Based on these risk profiles, the Mean Absolute Percentage Error (MAPE) and Mean Percentage Error (MPE) has been calculated to evaluate the expected accuracy of the time and cost predictions, EAC(t) and EAC, respectively.

Simulated EAC(t) accuracy		
method - PF	MAPE [%]	MPE [%]
PV - 1	2.86	0.82
PV - SPI	3.41	1.42
PV - SCI	3.60	1.22
ED - 1	2.02	-0.64
ED - SPI	3.56	1.26
ED - SCI	3.59	1.23
ES - 1	3.31	2.16
ES - SPI(t)	8.36	7.15
ES - SCI(t)	8.34	7.13

Simulated EAC accuracy		
method (PF)	MAPE [%]	MPE [%]
1	0.44	0.44
CPI	0.45	0.45
SPI	0.44	0.44
SPI(t)	1.03	1.03
SCI	2.24	2.21
SCI(t)	1.03	1.03
0.8 CPI + 0.2 SPI	2.23	2.20
0.8 CPI + 0.2 SPI(t)	0.58	0.58

According to the MAPE values¹ the best performance for time forecasting can be expected from the unweighted Earned Schedule method. For cost forecasting the unweighted and CPI-weighted methods should yield the best results.

2.3.2. Tracking description

Tracking authenticity

Manual tracking was performed over 21 tracking periods. The Real Duration and Real Cost mentioned in section “2.3.3. Earned Value Management” are based on manual user input.

The tracking information obtained from the project owner and introduced in ProTrack includes actual activity start dates, durations and costs.

¹ The MAPE gives the best indication for the forecast accuracy (the lower the MAPE, the more accurate the method) since all deviations from the targeted real duration (real cost) are cumulated, whereas for the MPE underestimates can be compensated by overestimates and vice versa, possibly leading to an overly positive evaluation of a certain method. However, the MPE can provide useful information about the nature of the deviations, i.e. does the method rather underestimate or overestimate the real duration (real cost)?

2.3.3. Earned Value Management

2.3.3.1. Performance metrics

	CV [€]	SV [€]	SV(t) [d]	CPI [-]	SPI [-]	SPI(t) [-]	p-factor [-]
avg	-22796.29	-8541.52	-3.48	1.00	1.00	0.99	1.00
std dev	33543.31	10348.65	5.01	0.00	0.00	0.01	0.00
final	-103475	0.00	-20.25	0.99	1	0.95	1

2.3.3.2. Time forecasting

PD	360	Real Duration	375	Late/early	3,98%
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EAC(t)	Real Accuracy			
method - PF	avg [d]	std dev [d]	MAPE [%]	MPE [%]
PV - 1	360.57	0.51	3.68	3.68
PV - SPI	360.62	0.50	3.66	3.66
PV - SCI	361.19	1.21	3.51	3.51
ED - 1	361.33	3.85	3.57	3.47
ED - SPI	361.48	3.82	3.53	3.43
ED - SCI	361.48	3.82	3.53	3.43
ES - 1	363.57	4.40	2.97	2.87
ES - SPI(t)	364.76	4.11	2.65	2.56
ES - SCI(t)	364.76	4.11	2.65	2.56

2.3.3.3. Cost forecasting

BAC	14,362,625€	Real Cost	14,466,100 €	Over/under budget	0.72%
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EAC	Real Accuracy			
method (PF)	avg [€]	std dev [€]	MAPE [%]	MPE [%]
1	14385421.29	33543.31	0.56	0.56
CPI	14385800.71	33544.91	0.56	0.56
SPI	14385721.29	33474.16	0.56	0.56
SPI(t)	14404382.62	33352.64	0.43	0.43
SCI	14386101	33477.17	0.55	0.55
SCI(t)	14404766.19	33290.60	0.43	0.42
0.8 CPI + 0.2 SPI	14385784.81	33530.64	0.56	0.56
0.8 CPI + 0.2 SPI(t)	14389491.48	31593.18	0.53	0.53